Speaker 1: Charin Phuaphumcharoen 6730300116

Linux Process Management Monitor

System-Level Resource Tracking

Speaker 1: Charin Phuaphumcharoen 6730300116

Project Objectives



Kernel OS

Deep architectural understanding.



Virtual File System

Direct /proc extraction.



System Calls

Low-level process execution.

Speaker 2: Krairawee Boonthad 6730300051

System Architecture

User Space

C++ Application.

ncurses Interface.



Kernel Space

/proc Filesystem.

POSIX Signals.

Speaker 2: Krairawee Boonthad 6730300051

Data Extraction



/proc/[pid]/stat

Execution state and CPU ticks.



/proc/[pid]/status

VmRSS physical memory allocation.

Speaker 3: Siwakorn Pratumsuwan 6730300574

Core Logic

CPU Percentage

Process time delta versus System delta.

Process Hierarchy

Parent-Child PID mapping logic.

Speaker 3: Siwakorn Pratumsuwan 6730300574

Terminal Interface



ncurses Library

Structured terminal rendering.



Real-Time Updates

Zero-flicker dashboard refresh.

Speaker 4: Kittikanoot Sangiam 6730300701

Process Control

SIGSTOP

Pause execution.

SIGCONT

Resume execution.

SIGKILL

Force termination.

Speaker 4: Kittikanoot Sangiam 6730300701

The Stubborn Process

Dummy process catching SIGTERM.

Requires SIGKILL kernel override.

```
void handle_sigterm(int sig) {
        cout << "I refuse to die!";
}

int main() {
        signal(SIGTERM, handle_sigterm);
        while(true) { }
}
```

Speaker 5: Soungwut Konak 6730300175

Live Demonstration



Interactive System Tracking

Deploying signals in real-time.

Speaker 5: Soungwut Konak 6730300175

Conclusion

Thank you for your attention.



GITHUB REPOSITORY